

Davide Capone

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SUMMARY

AI / Machine Learning Engineer with professional experience in deep learning and computer vision, focused on industrial applications. Strong background in Python, PyTorch, and TensorFlow, with hands-on work in object detection, object tracking, and model deployment. Experienced in collaborating with engineering and business teams to deliver reliable, data-driven AI solutions.

EXPERIENCE

Data Scientist

Danieli S.p.A.

Jul 2025 – Present

Buttrio (UD), Italy

- Built and evaluated machine learning models for business forecasting and scenario analysis, supporting data-driven planning and performance monitoring.
- Designed and implemented a machine learning-based computer vision system, including data preprocessing, model training, evaluation, and deployment, to support industrial decision-making processes.
- Developed data-driven analytical reports and interactive dashboards to support business intelligence and operational decision-making using SQL and Power BI.

Machine Learning Engineer (Intern)

Danieli Automation S.p.A.

Dec 2024 – May 2025

Buttrio (UD), Italy

- Designed and developed an end-to-end multi-object tracking system for industrial steel production environments, addressing real-world operational constraints.
- Evaluated multiple state-of-the-art deep learning object detection models to identify optimal trade-offs between accuracy, robustness, and inference performance.
- Integrated selected detection models with Kalman filtering and data association techniques to implement a robust multi-object tracking pipeline.
- Optimized and deployed trained models in ONNX format to enable efficient and scalable industrial inference.

Machine Learning Engineer (Intern)

Danieli Automation S.p.A.

Oct 2022 – Dec 2022

Buttrio (UD), Italy

- Applied advanced computer vision techniques, including camera calibration and homography estimation, to enable accurate object detection and spatial localization in industrial environments.
- Designed, trained, and evaluated convolutional neural network (CNN) models for object detection and localization under real-world industrial constraints.
- Conducted model evaluation, hyperparameter tuning, and performance analysis to identify optimal solutions for potential production deployment.

EDUCATION

University of Trieste

Master of Science in Data Science and Scientific Computing

Trieste, Italy

Aug 2022 – May 2025

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, Optimization

University of Udine

Bachelor of Science in Internet of Things, Big Data, and Machine Learning

Udine, Italy

Aug 2019 – May 2022

- Relevant coursework: Algorithms and Data Structures, Machine Learning, Statistics, Cloud Computing

PROJECTS

GAPC – Asteroid Observation Catalog

Web Application Project

Python, Django, Databases

- Developed a Django application to catalog and manage asteroid observations following astronomical data standards.
- Designed database schemas and backend logic to efficiently store, retrieve, and analyze scientific metadata.
- Implemented data management features to support structured analysis and research workflows.

YOH024 – Sound Event Detection Model

Deep Learning Project

Python, PyTorch, Audio Processing

- Designed and implemented a deep learning model for sound event detection inspired by the YOLO (You-Only-Hear-Once) paradigm.
- Extended the original YOHO architecture by integrating improvements from later YOLO versions to enhance efficiency and detection accuracy.
- Evaluated model performance on standard audio segmentation datasets and benchmarked results against the original YOHO implementation.

Weighted Matrix Factorization Recommender

Machine Learning Project

Python, Recommender Systems

- Developed a movie recommendation system based on weighted matrix factorization techniques.
- Implemented and compared optimization strategies including Stochastic Gradient Descent (SGD) and Weighted Alternating Least Squares (WALS).
- Analyzed model performance to improve recommendation accuracy and system efficiency.

Traffic Lights Optimization

Reinforcement Learning Project

Python, Reinforcement Learning, Simulation

- Applied reinforcement learning methods, including Value Iteration and Policy Iteration, to optimize traffic light control policies.
- Modeled traffic environments and simulated agent–environment interactions to reduce queue lengths and waiting times.
- Evaluated learned policies through simulation-based performance analysis.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Polars, OpenCV, Django

MLOps & Deployment: Model Evaluation, Hyperparameter Tuning, ONNX Export, Inference Optimization

Data & Analytics: Feature Engineering, Data Preprocessing, Exploratory Data Analysis (EDA), Business Intelligence

Tools: Git, Docker, Power BI, Databases

CERTIFICATIONS

- Introduction to Deep Learning & Neural Networks with Keras — IBM (TensorFlow, Keras, Neural Networks)
- Machine Learning in Production — DeepLearning.AI (MLOps, Deployment, Monitoring)
- Machine Learning with Python — IBM (Python, Scikit-learn, Feature Engineering)